

COURSE TITLE : PRESENT PRINTING INDUSTRY
COURSE CODE : 6105
COURSE CATEGORY : E
PERIODS/ WEEK : 5
PERIODS/ SEMESTER : 75
CREDIT : 5

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Introduction to present printing industry	15
2	Common pollution prevention practice in printing	25
3	present printing scenario in lithography and screen printing	20
4	Present printing scenario in gravure and flexography printing	15
TOTAL		75

Course Outcomes :

G .O.	ON THE COMPLETION OF THE STUDY OF THIS MODULE STUDENT WILL BE ABLE :
1	To understand about waste management in printing industry
2	To understand Common pollution prevention practice in printing
3	To know present printing scenario in lithography and screen printing
4	To know the Present printing scenario in gravure and flexography printing

SPECIFIC OUTCOMES

MODULE I INTRODUCTION TO PRESENT PRINTING INDUSTRY

1.1.0. To understand about waste management in printing industry

- 1.1.1. To Explain Pollution Prevention and Control
- 1.1.2. To specify different Treatment Technologies
- 1.1.3. To know printing industry Emissions Guidelines

MODULE II COMMON POLLUTION PREVENTION PRACTICE IN PRINTING

2.1.0 To understand Common pollution prevention practice in printing

- 2.1.1 To discuss how to reduce waste
- 2.1.2 To state job planning
- 2.1.3 To explain waste segregation, recycling and reuse

MODULE III PRESENT PRINTING SCENARIO IN LITHOGRAPHY AND SCREEN PRINTING

3.1.0 To know present printing scenario in lithography and screen printing

- 3.1.1 To discuss the overview to lithographic printing
- 3.1.2 To describe the changes in prepress, press and post press
- 3.1.3 To discuss the overview to screen printing

- 3.1.4 To describe the changes in prepress, press and post press

MODULE IV PRESENT PRINTING SCENARIO IN GRAVURE AND FLEXOGRAPHY PRINTING

4.1.0 To know the Present printing scenario in gravure and flexography printing

- 4.1.1 To discuss an overview to gravure printing
- 4.1.2 To describe the changes in prepress, press and post press
- 4.1.3 To discuss an overview flexographic printing
- 4.1.4 To describe the changes in prepress, press and post press

CONTENT DETAILS

MODULE I

Introduction to present printing industry

Different products – waste characteristics – emission to air – discharge to water bodies –noise – waste with environmental hazardous – pollution prevention control, industry guidelines. Treatment technologies. Emission guidelines – monitoring and reporting.

MODULE II

Common pollution prevention practice in printing

Major Waste streams - How to Reduce Waste Start at the Beginning-Graphic Design – Job Planning - Material Handling and Storage - Waste Segregation, Recycling, and Reuse - Alternative Materials – Solvents – Prepress , Photochemistry - Silver Recovery – Alternative Prepress Technologies – Press ,Mechanical Modifications – Inks , Recycling Inks - Alternative Inks - Ultraviolet (UV) - Electron Beam Curable (EBC) - Vegetable Oil-Based Inks - Water-Based Inks - Waterless Inks – Cleanup , Parts Washers. Post press.

MODULE III

Present printing scenario in lithography and screen printing

Overview. Prepress - Photo Processing – Proofs – Plates. Press - Materials Substitution -Inks - Fountain Solution - Cleaning Solutions - Parts Washing. Post Press. Screen Printing – Overview. Prepress, Stencils and Screen Preparation. Press – Ink. Post press, Screen Reclamation - Equipment Modification – Process -Modifications - Screen Disposal. Automation in lithography and screen printing.

MODULE IV

Present printing scenario in gravure and flexography printing

Flexography- Overview – Prepress - Plate Preparation – Press, Inks - Post Press Letterpress. Gravure, Overview – Prepress- Cylinder Preparation – Press - Process Modification – Post Press - Equipment Modifications - Process Modification. Automation in gravure and flexography printing.

Reference :		
Author	Title	Publishers
Helmut Kipphan	Hand book of print media	Springer Science & Business Media
HL Apfelberg and MJ Apfelberg	Implementing quality management in graphic arts	GATF
Howard M. Fentan	On demand Digital Printing	GATF

